

Ashok Prasad Neupane

Machine Learning Researcher | AI Engineer

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Portfolio: ashokpneupane.com.np | Google Scholar

Summary

ML researcher working on RL post-training of diffusion models, agentic RL, and streaming vision-language models; first/co-first author on 3 papers (CVPR 2026 Workshop; under review at ICLR, WACV, TMLR). Founding engineer (1 of 2) at Elephant Intelligence, building and benchmarking self-improving browser-use agents. Interests: RL for generative models and LLM agents, controllable 3D generation, efficient VLM inference.

Education

Pulchowk Campus, IOE, Tribhuvan University

Bachelor of Engineering in Computer Engineering

St. Xavier's College

High School (+2), Science

2022 – 2026

Ranked 47 of 15,000+ in national entrance exam

2019 – 2021

Kathmandu, Nepal

Experience

Elephant Intelligence (stealth-stage) | elephantintelligence.ai ↗

San Francisco, CA (Remote)

Founding AI Engineer (1 of 2) – Browser-Use Agents

Jun 2026 – Present

Worked directly with founders who bring 10+ years of ML research and production AI experience across Apple, Amazon, Microsoft Research, Roche, Qualcomm, SRI, and PathAI.

- Building an AI agent harness for browser-use agents specializing in web **form filling and navigation**; as one of two founding engineers, owned both the core product and its evaluation stack end to end.
- Built an in-house **browser-agent evaluation benchmark** on **BrowserGym**, a Gym-style RL environment where web agents act and are evaluated, to measure success rate, latency, token usage, and cost against competing products.
- Drove task success rate from **~20% to 100%** on the benchmark; the agent now runs **~2× faster** and **25%+ cheaper than the closest competitor** – currently the most accurate, fastest, and cheapest solution in the space (ahead of public launch).
- Shipped the product as a **desktop app, REST API, and browser extension**; deployed to beta testers, prospective customers, and investors, and integrated into third-party software; managed deployment on **AWS** and other cloud infrastructure.
- Built an **LLM routing** system that sends each request to the cheapest model able to answer it accurately; developed novel routing ideas that beat recent published methods by **10%** and further lowered product cost.
- Cut our own per-task cost by **4–5×** with **continual learning** approaches that replace repeated LLM calls on every step.

Sceneverse | sceneverse.ai ↗

Jan 2025 – May 2025

Machine Learning Engineer

Remote

Worked alongside world-class professors, research scientists, and PhD researchers from INSAIT, ETH Zurich, and Meta.

- Built 3D generative AI training and inference workflows for controllable, realistic indoor scene synthesis.
- Developed synthetic data pipelines for photorealistic 3D asset rendering, dataset preparation, and validation, inspired by Apple's Hypersim.
- Trained and evaluated 3D scene foundation models on multi-GPU compute with experiment tracking.

Freelance AI Engineer

Jan 2024 – Present

Independent, client-based

Remote

- Delivered an end-to-end **text-to-video** pipeline: LLM script generation, diffusion-based lip sync, audio-visual alignment, and voice cloning.
- Built a **RAG-based grading agent** for schools in Kathmandu Valley using document retrieval, LLM reasoning, and prompt evaluation.
- Built a lightweight **VLM-based deepfake detector** reaching **98.7%** accuracy on seen and **87.3%** on unseen generators.

Fusemachines | fusemachines.com ↗

Dec 2022 – Feb 2024

AI Fellow

Kathmandu, Nepal

- Completed a 15-month AI fellowship spanning ML, deep learning, CV, NLP, and production ML; implemented and deployed PyTorch/TensorFlow models behind APIs.

Publications

iARCS: Iterative Agentic RL for Controllable 3D Scene Generation ↗

CVPR 2026 Workshop

Co-first author; extended version under review at WACV ▶ [Project Page](#) ▶ [Paper \(arXiv\)](#) ▶ [Code](#) ▶ [Dataset](#)

- Agentic RL framework adapting a pretrained 3D scene diffusion model to natural-language task constraints: universal-reward pretraining, then DDPO + LoRA fine-tuning on **LLM-written reward programs** iteratively refined via a reflection loop.
- Raised task success from **4.5% to 73.8%** (robot reachability), **5.2% to 66.0%** (TV visibility), and **0.9% to 33.4%** (study zone) over MiDiffusion; beats ATISS, InstructScene, PhyScene on physical plausibility.
- Released 12K generated scenes; augmenting training data with iARCS scenes cut MiDiffusion object collisions **52.7%→41.5%**. Preferred over real 3D-FRONT scenes by humans 61% of the time.

Foresight: Planning Future Perception in Streaming VLMs without Retraining ↗

Under review, ICLR

First & corresponding author ▶ [Project Page](#) ▶ [Code](#)

- Training-free dual-stream architecture for streaming VLMs: two Siamese LLMs sharing weights and KV cache, one ingesting frames continuously while the other plans future perception (when to reason, what to check, sampling FPS) via schema-guided decoding and diff-based updates.
- With a frozen Qwen3-VL-8B, achieved **23.0 joint F1** on OmniPro Online, beating the strongest *trained* baseline by **9.5%**; improved the backbone by **+6.7** on StreamingBench and **+15.4** on OVO-Bench with zero training.

iADD: Improving Alignment and Diversity in Diffusion Policy Optimization ↗

Under review, TMLR

Co-first author ▶ [Project Page](#) ▶ [Code](#)

- Proved that late-timestep-only RL updates harm diversity (contrary to prior work) and proposed incremental-timestep training with **Feynman-Kac particle guidance** for RL post-training of diffusion models.
- Best alignment-diversity trade-off across text-to-image (SD v1.4), 3D scenes, and vanishing-point tasks vs. DDPO and B²-DiffuRL: rare-prompt success **24.8%→66.9%**, reward-diversity AUC **53.8%→77.7%**, vanishing-point rarity **31.1%→95.1%**; iADD+GRPO beats DanceGRPO and BranchGRPO.

Teaching Experience

Mid-Valley International College | [midvalley.edu.np](#) ↗

Maitidevi, Kathmandu

Instructor – Data Science; AI & Machine Learning (Bachelor's level)

- Taught two undergraduate courses, *Data Science* and *AI & Machine Learning*, covering data analysis, classical ML, deep learning, and model evaluation.

Skill Shikshya | [skillshikshya.com](#) ↗

Oct 2024 – Jan 2025

AI & Programming Instructor

Kathmandu, Nepal

- Delivered 255+ hours of Python, prompt engineering, agentic AI, and RAG training to individual learners and professional cohorts at a leading IT training institute in Nepal.

Projects

Fashion Virtual Try-On ↗ | Diffusion, Computer Vision ▶ [Code](#)

- Fine-tuned Improved Distribution Matching Distillation diffusion models for **20× faster** try-on without fidelity loss; benchmarked CatVTON, IDM-VTON, OOTDiffusion. KU Hackfest AI Track Winner; Hult Prize finalist; presented at Locus Research Symposium.

Human Action Recognition (with Khalti) ↗ | Video Transformers ▶ [Demo Video](#) ▶ [Code](#)

- VideoMAE-based model with **95%+** accuracy on Nepali video data; designed log-linear sequence scaling for efficient long-video inference.

Technical Skills

ML / DL: PyTorch, TensorFlow, Hugging Face Transformers, Diffusers, LoRA/PEFT, Diffusion Models, VLMs, BERT, CV, NLP

LLMs & Agents: LLM Agents, Browser-Use Agents, Agent Harnesses, LLM Routing, Tool Use, RAG, Embeddings, Vector Search, LangChain, Prompt Engineering, OpenAI/Gemini APIs, LLM Evaluation & Benchmarking

RL & Post-training: RLHF, DDPO, GRPO, Reward Modeling, Fine-tuning, Inference Optimization, KV-Cache Management

Engineering: Python, C/C++, SQL, PostgreSQL, NoSQL, FastAPI, REST APIs, React, Three.js, Docker, Linux, Git

MLOps & Cloud: Data Pipelines, Synthetic Data, CUDA/GPU, HPC/Multi-GPU Training, Model Serving, AWS, Azure

Awards & Honors

- **Samsung Innovation Campus** – Top 30 nationally (Coding & Programming; Big Data Computing tracks)
- **Generative AI & LLMs Course** – Selected among Top 20 Pulchowk Campus students
- **Hackathons:** AI Track Winner, KU Hackfest (Nov 2024); Best AI Product, Hackademia 2 (Sep 2024); Best Presentation, Hackademia (Dec 2023); Geek-A-Thon (Aug 2023); Engen Hacks (Oct 2023); Civil Engineering Hackathon (Dec 2024); Leo Club Ideathon (Dec 2024)
- **Model Competitions:** BE Model Winner, Pulchowk Campus (Jan 2024); All Nepal Civil Engineering Model Competition Winner (May 2025)